

Downs, Distance, and Field Position

Four tries to go ten yards, the yard line as money, and the expected-points scorecard that prices every snap.

The graphic in the corner of the screen changes from **2nd & 3** to **3rd & 8**. Nothing else has happened yet: the offense is still in its huddle. But the announcer is already calling it “an obvious passing down,” the defense is sending a sixth defensive back onto the field, and the crowd is on its feet because the home team is on defense. Two plays earlier the same graphic read **1st & 10** and nobody in the building knew what was coming. How can two numbers in the corner of the screen tell 70,000 people so much? This chapter answers that, and then goes one step further: it gives you a way to put a price on every snap, so that “he gained eight yards” stops sounding automatically like good news.

You'll be able to...

- Explain the four-down system and the line to gain, and say why teams usually punt on fourth down instead of going for it.
- Read the down-and-distance graphic, sort it into short, medium or long, and say what each bucket does to the offense's options.
- Point to the line of scrimmage, explain how the officials spot the ball, and describe what the chain crew (the sticks) does now that cameras do the measuring.
- Name the ways the ball changes hands (interception, fumble, turnover on downs, punt) and say why a sack hurts more than the yards it costs.
- Treat field position as money: say roughly what a first down at your own 25, at midfield and at the opponent's 20 is worth in expected points.
- Read EPA and success rate as a per-play scorecard: what a +2.0 or a -0.5 means, and why a gain can score as a loss.

Four downs to go ten yards

Here is the whole system in two sentences. The offense gets **four tries to gain ten yards**. Make it, and it gets four fresh tries; fail, and the ball goes to the other team.

Each try is a **down**: one play, counted from 1 to 4. The yard line ten yards ahead of where the series began is the **line to gain**, which on television is the yellow line painted across the picture. Reach it and you have earned a **first down**: the count resets to 1, and a new line to gain is set ten yards farther on. Announcers use “first down” both for the first play of a series and for the act of earning a new one (“they picked up the first down”).

The second number on the graphic is the **distance**: how many yards are still needed to reach the line to gain. “2nd & 6” means it's the second down and six yards remain. Watch a single series play out:

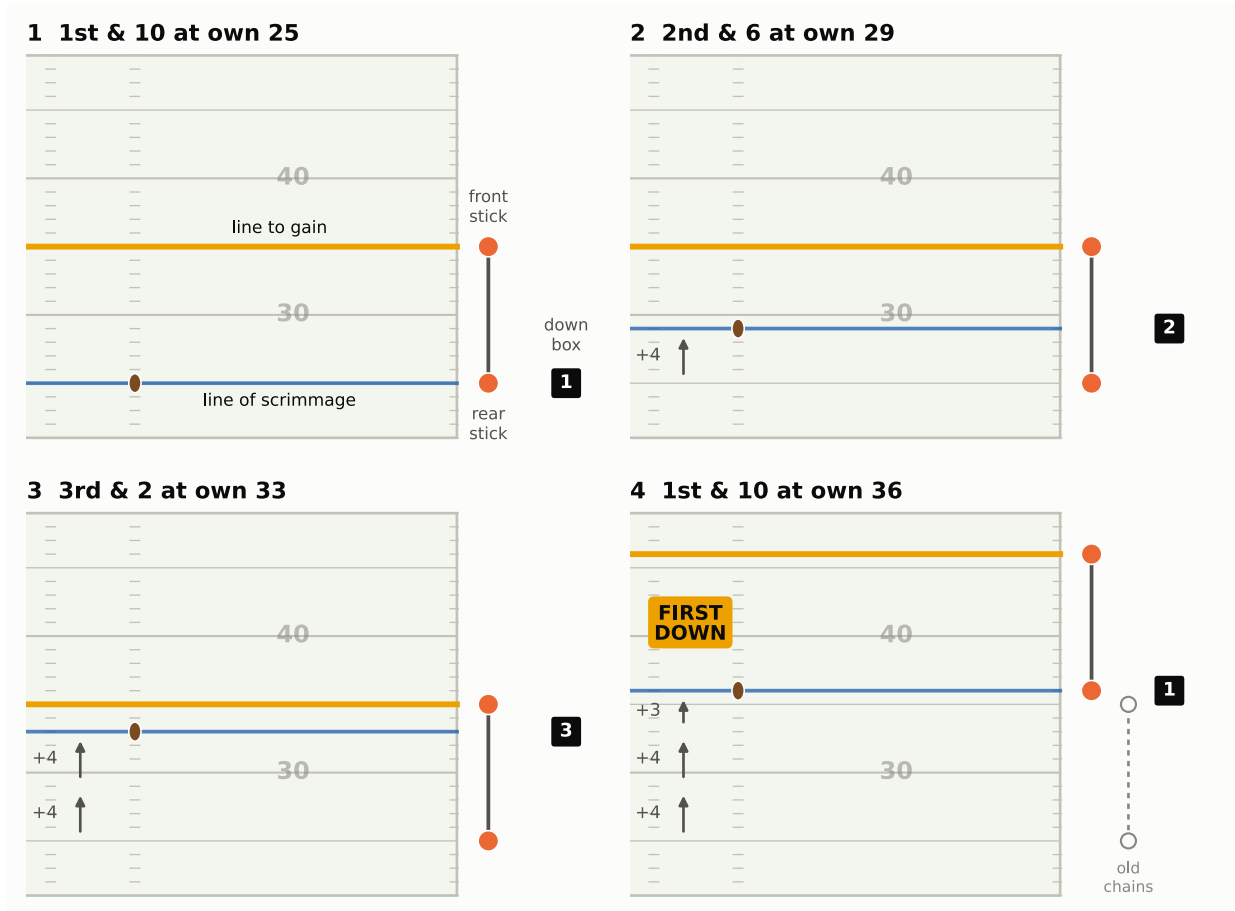


Figure 1: One series of downs, offense moving up the page, the ball on the right hash. (1) First-and-10 at the offense's own 25: the line to gain (yellow) is the 35, marked on the sideline by the chain crew's front stick; the rear stick marks where the series began, and the down box shows which down it is. (2) A 4-yard run makes it second-and-6. (3) A 4-yard pass makes it third-and-2. (4) A 3-yard run crosses the line to gain: first down. The chain crew picks up the sticks and resets them, rear stick at the new spot, front stick ten yards on.

Notice what *doesn't* reset. The yard line keeps advancing play by play; only the down count and the line to gain start over when the offense earns a first down. A series can also go backward: lose three yards on first down and it's "2nd & 13", because the line to gain hasn't moved and the ball has. Penalties do the same: a five-yard penalty on the offense on first down makes it "1st & 15."

How often does an offense keep the chain moving? In the 2025 regular season, 71% of fresh sets of downs ended in another first down or a touchdown.¹ That number is the hidden engine of the whole game. A drive is a chain of these little four-down contests; win roughly seven of every ten and you will usually string together enough of them to get near the end zone, lose one and the drive is over.

¹Course calculation from nflverse play-by-play ([series_success](#)), 2025 regular season: every series of downs, counted once; a series succeeds if it ends in a first down or a touchdown.

! Why it exists: the rule that made the game move

American football grew out of rugby, and in its earliest college form there was no requirement to gain ground. A team could keep the ball as long as it didn't lose it. Princeton and Yale played scoreless ties in 1880 and 1881, and in the second, remembered as the "block game," Princeton held the ball for the whole first half without trying to advance it and Yale did the same in the second, because a tie was enough.² In 1882 Yale's **Walter Camp**, a halfback (and by then a medical student) who had already made himself the sport's most influential rule-maker, proposed a fix: keep the ball only if you advance five yards in three downs. His fellow committee members thought it unworkable and adopted it only on condition that it could be dropped if it failed. It didn't. The distance was doubled to ten yards in 1906, the same year the forward pass was legalized, and a fourth down was added in 1912, which gave the modern "four downs to go ten."³ The lines chalked across the field every five yards so that officials could judge the distance are the usual explanation for the field's nickname, the *gridiron*, though some historians credit the lengthwise lines added in 1903, which briefly turned the field into a true checkerboard.⁴

The trade the rule makes is the heart of the sport: the offense gets a **guarantee** (four tries) in exchange for an **obligation** (make progress, or give the ball back). Every strategic idea in this book, from third-down defenses to fourth-down analytics, lives inside that bargain.

Reading the graphic

Every broadcast keeps a running summary at the top or bottom of the screen, the **score bug**. Networks style it differently, but the pieces are always the same:

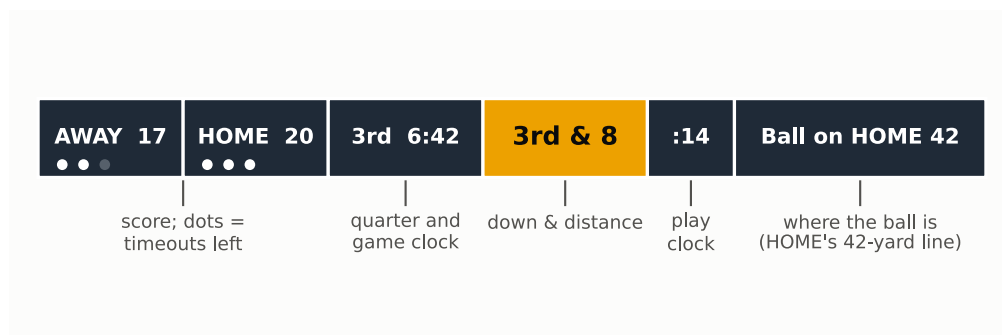


Figure 2: A generic score bug (every network styles its own). The down and distance is the box to read first: here, third down with 8 yards to go. The play clock counts down to the snap; the dots under each team are timeouts left in the half. 'Ball on HOME 42' means the ball is on the home team's 42-yard line, so the visiting offense is 42 yards from a touchdown.

The score, the clock and the timeouts were the subject of *The Game in Fifteen Minutes*. The new piece is the yellow box, and two details about it are worth knowing.

'& Goal.' When the line to gain would fall in or past the end zone, the graphic says **1st & Goal** instead of a number: the only way to reach the line to gain is to score. From the opponent's 6, it's 1st & Goal, and a three-yard gain makes it 2nd & Goal from the 3. There are no more first downs available, only the goal line.

'& Inches.' When the ball is spotted just short of the line to gain, broadcasters say "4th & inches." The graphic usually rounds it to 1.

²The 1880 game (Polo Grounds, Nov. 25, 1880, 0–0, played in frozen conditions): Action Football Sundays, "1880's Game of the Season": <https://actionfootball Sundays.substack.com/p/1880s-game-of-the-season>. The 1881 "block game" (Polo Grounds, Nov. 24, 1881, 0–0; each team held the ball for a half): Pigskin Dispatch, "Football History Rewind: 1881": <https://pigskindispatch.com/1881-football-history/>; "1881 Princeton Tigers football team," Wikipedia: https://en.wikipedia.org/wiki/1881_Princeton_Tigers_football_team

³*The Saturday Evening Post*, "The Quiet Yalie Who Invented American Football" (Sept. 2021), on Camp's 1882 downs-and-distance rule (five yards in three downs, adopted on condition it could be dropped), the 1906 change to ten yards and the 1912 fourth down: <https://www.saturdayeveningpost.com/2021/09/the-quiet-yalie-who-invented-american-football/>; *Encyclopedia.com*, "Camp, Walter Chauncey": https://www.encyclopedia.com/topic/Walter_Chauncey_Camp.aspx; "1906 college football season," Wikipedia (first-down distance doubled to ten yards in three downs, forward pass legalized): https://en.wikipedia.org/wiki/1906_college_football_season; Pro Football Hall of Fame, "Changing the Rules" (1912: four downs to gain ten yards): <https://profootballhof.com/news/2011/12/news-changing-the-rules-archived2>. Camp played halfback for Yale from 1876 to 1882 ("Walter Camp," Wikipedia: https://en.wikipedia.org/wiki/Walter_Camp).

⁴The five-yard lines date from Camp's 1882 downs rule; the lengthwise lines were added for 1903 and dropped by about 1920. Pigskin Dispatch, "History of Field Markings" (crediting the checkerboard and a 1911 use of the name): <https://pigskindispatch.com/home/Football-Fun-Facts/History-of-Field-markings>; "Gridiron football," Wikipedia (the name from the grid of lines along both width and length): https://en.wikipedia.org/wiki/Gridiron_football

Short, medium and long

Coaches, broadcasters and this book sort distances into three buckets, because each one changes what the offense can realistically do:

- **Short** (1–3 yards): a good run gets it. The offense has its whole menu, runs and passes, and the defense has to respect both.
- **Medium** (4–6 yards): a run *can* get it but usually doesn't; most offenses throw short passes that are designed to get just past the line to gain.
- **Long** (7 yards or more): only a pass, or a rare big run, gets there. The defense knows it and drops more players into coverage.

The buckets matter most on third down, because that's usually the last real chance before the offense has to punt. Here is how often third downs were converted over the last ten seasons, by the yards needed:

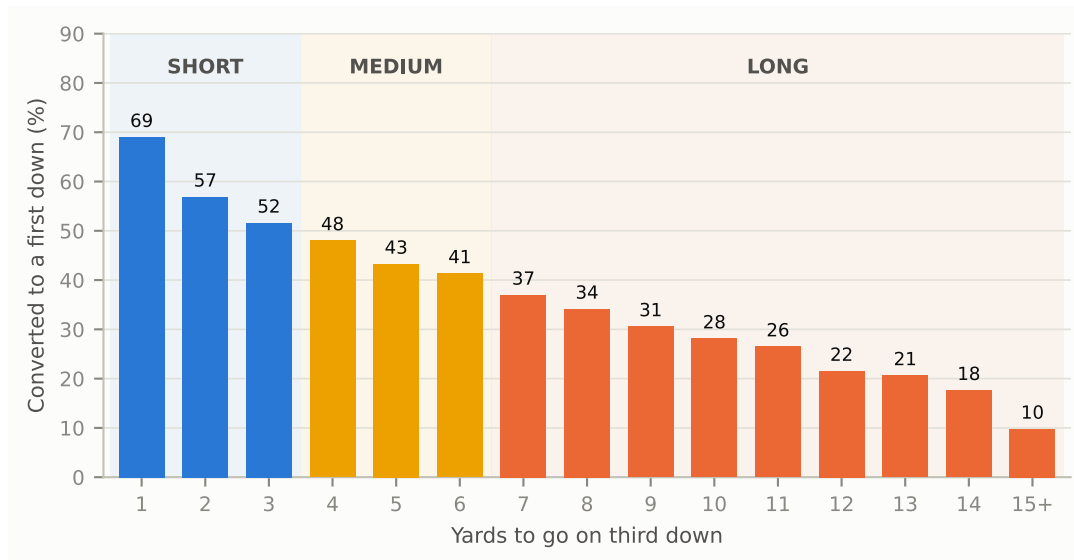


Figure 3: Third-down conversion rate by yards to go, every third-down run and pass in the 2016-2025 regular seasons (sacks and scrambles count as passes). Each extra yard costs a few percentage points. Third-and-1 is converted about seven times in ten; by third-and-10 it's under three in ten. Source: nflverse play-by-play.

Read the chart as a price list. Every yard you leave for third down costs you a few percentage points of keeping the drive alive. That is why coaches talk about first and second down as setting up third: a four-yard gain on first down doesn't look like much, but it turns a likely third-and-long into a manageable third-and-short. The language coaches use for this, staying "on schedule," is built up in Early Downs and Field Zones, and third down gets its own chapter, Third Down: The Money Down.

What the graphic tells you about the next play

The down and distance doesn't just predict whether the offense will convert. It predicts **what it will try**. Here is how often offenses called a pass in each situation in 2025, leaving out the end of each half and lopsided games, where the score and the clock dictate the call:

	1-3 yds (short)	4-6 yds (medium)	7-10 yds (long)	11+ yds (very long)
1st down	rare	rare	46% 7,445 plays	69% 318 plays
2nd down	30% 989 plays	48% 1,543 plays	69% 2,710 plays	82% 891 plays
3rd down	46% 1,165 plays	93% 914 plays	97% 1,072 plays	92% 681 plays

Figure 4: How often the offense passed, by down and distance, 2025 regular season, neutral situations (outside the last two minutes of each half, win probability 20-80%). ‘Pass’ means a called pass, so sacks and scrambles count. First down is almost always first-and-10; the short first-down cells come from goal-line and penalty situations and are marked ‘rare’. Source: nflverse play-by-play.

Three things jump out, and each one has a reason.

1. **First-and-10 is the most balanced situation in football.** Offenses pass a little under half the time. Nothing about the situation forces their hand, so it’s where play-callers have the most freedom, and where defenses know the least.
2. **Third-and-medium or longer is close to a sure pass.** Close to nine times in ten on third-and-4, and above 90% at every longer distance. A run gains four yards less than half the time, so on the down that decides whether you punt, offenses throw. That predictability is the defense’s best friend: it can bring extra pass rushers or extra defensive backs without worrying much about the run.
3. **Short yardage is a real toss-up.** On third-and-1 to 3, offenses still threw 46% of the time, because a defense that crowds the line to stop the run is a defense that leaves receivers alone.

That grid is the beginning of reading the game. Before you know anything about formations or personnel, the graphic already tells you whether the offense’s options are wide open or nearly gone.

🔥 Common misconception

“Second-and-1 is a running down.” It’s actually one of the best situations in football to throw deep. The offense can afford to waste a down: even if the pass falls incomplete, it’s third-and-1, still very likely to convert. Defenses know that too, so on second-and-short they often guard against the long pass, which in turn makes the run easier. Offenses called a pass on 30% of second-and-short snaps in 2025; many of those were play-action shots (a fake handoff followed by a throw downfield, built in Play-Action, Bootlegs, and Screens).

Where the ball is: the line of scrimmage and the spot

Every play starts from a line. The **line of scrimmage** is the imaginary line across the field through the ball before the snap; each team must stay on its own side of it until the ball moves. (Strictly speaking each team has its own line through its end of the ball, and the ball’s length between them is a no-man’s-land called the neutral zone, which is why you hear “neutral zone infraction” penalties; see The Pre-Snap Phase.) On a broadcast it’s the blue line the network paints across the picture. The snap starts the play from there.

Where does the next line of scrimmage come from? From the **spot**: where the officials place the ball after the previous play. The rule behind it is **forward progress**: the ball is spotted at the farthest point it reached toward the opponent’s goal before the play ended, not where the ball carrier ended up after being driven backward (forward progress is one of the rules of the ball covered in Part 9). Then it’s moved sideways, if necessary, to the nearer row of hash marks, so every play starts between them.

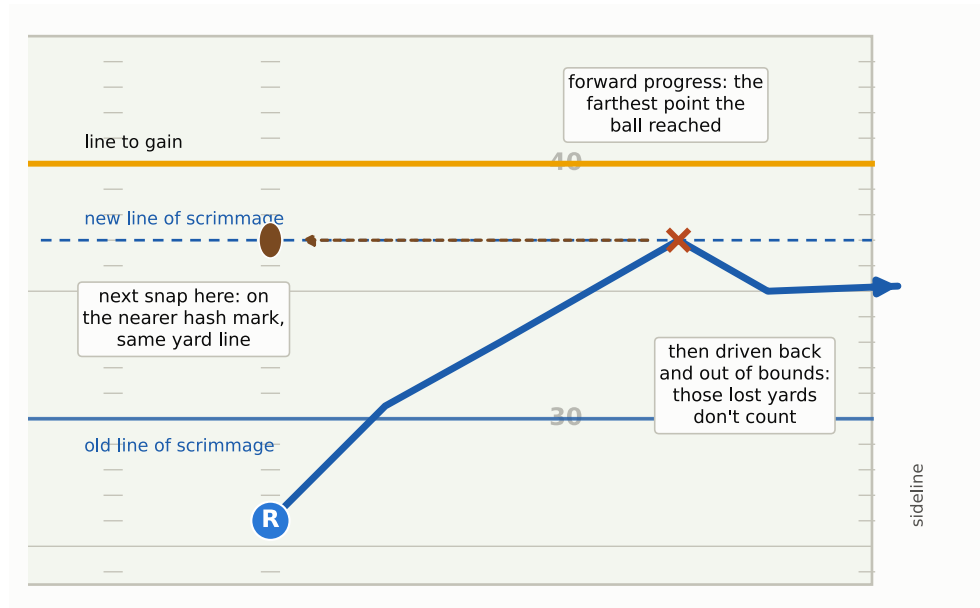


Figure 5: How the next line of scrimmage is set. The runner gets as far as the X (his forward progress), is driven back two yards and pushed out of bounds. The officials spot the ball at the X's yard line, not where he went out, and bring it in to the nearer hash mark. That spot becomes the new line of scrimmage (dashed); the line to gain hasn't moved.

The spot matters more than its size suggests. On third-and-1 the difference between a good spot and a bad one is the difference between a first down and a punt, and the spot is a **human judgment**, made by officials running alongside the play, often from the far side of a pile of bodies. That's why you see so many replays of a ball carrier's knee and the ball at the moment he went down.

i Madden vs. real life

In a video game the ball is spotted to the inch, instantly, and the yellow line on your screen is the line to gain. In the NFL, a person decides where the ball goes, by eye, and the yellow line you see on TV is the broadcaster's drawing of a line it was told about, not an official marking. When the spot and the line to gain are close, nobody (players, coaches or the TV truck) knows whether it's a first down until it's measured.

The sticks: the chain crew and the measurement

On one sideline stands a small crew holding two bright poles joined by a chain exactly ten yards long. These are **the sticks**, also called **the chains**. The **rear stick** marks where the series began; the **front stick** marks the line to gain. A third crew member holds the **down box**, a pole topped with a flip-card number, at the current line of scrimmage. You saw all three in the first figure.

When an offense earns a first down, the crew picks up the chain and resets it, rear stick at the new spot. That's where the phrase comes from: a team that keeps converting is **moving the sticks**, and you'll hear quarterbacks praised for "throwing to the sticks" on third down, meaning past the line to gain rather than short of it.

When the spot is close, the officials call for a **measurement**. For a century that meant the chain crew jogging onto the field, an official laying the rear stick at its mark (the crew clips the chain at a nearby five-yard line beforehand so it can be carried in without losing its place⁵), the chain pulled taut, and a referee bending over to see whether the front stick touches the tip of the ball. Football fans have watched it for generations: the crowd hushed, the referee holding up a hand to show the gap ("by inches").

That ritual is now mostly gone. **Starting with the 2025 season, the NFL measures first downs with cameras:** a Sony Hawk-Eye system of six 8K cameras in every stadium tracks the ball's spot and the line to gain and decides close calls in about 30 seconds, with a virtual replay shown on the broadcast and the stadium screens. The officials still *spot* the ball by eye; the cameras only measure from that spot. The chain crew is still on the sideline, as the visible reference for players and coaches and as a backup if

⁵"Chain crew," Wikipedia: a clip is attached to the chain at the rear edge of the five-yard line nearest the rear rod, so the chain can be carried out and laid back down exactly: https://en.wikipedia.org/wiki/Chain_crew

the system fails.⁶ The details of how officials spot the ball, and what replay can and can't change, are in Officiating, Replay, and the Rules of the Ball.

Fourth down: why teams usually punt

Now the logic of the fourth down. On fourth down the offense has three choices:

1. **Go for it:** run a normal play and try to reach the line to gain. Make it, and the series continues. Miss it, and the result is a **turnover on downs:** the other team takes over *right there*, wherever the play ended.
2. **Kick a field goal:** three points if it's good, but only realistic within a kicker's range (we'll come back to that line on the field).
3. **Punt:** a **punt** is a kick that hands the ball to the other team *on purpose*, as far down the field as possible. The punter stands about 15 yards behind the line of scrimmage, catches the long snap and kicks it out of his hands; the receiving team either returns it or signals a fair catch.

Why would any team give the ball away on purpose? Because it's going to lose the ball anyway if it fails on fourth down, and the punt lets it choose *where*. Fail on fourth-and-5 at your own 30 and the opponent starts its drive 30 yards from a touchdown. Punt from the same spot and, after a typical punt and return, the opponent starts somewhere around its own 25 or 30, 70-odd yards away. The punt trades possession, which you were probably going to lose, for roughly forty yards of field. The kicking game, including how punters and coverage teams fight for those yards, is in Field Goals, Extra Points, and the Punting Game.

In 2025 NFL teams punted on 51% of fourth downs, kicked a field goal on 26%, and went for it on 23%. But those averages hide the pattern that matters, which is that the choice depends on two things: how far the offense needs to go, and where it is on the field.

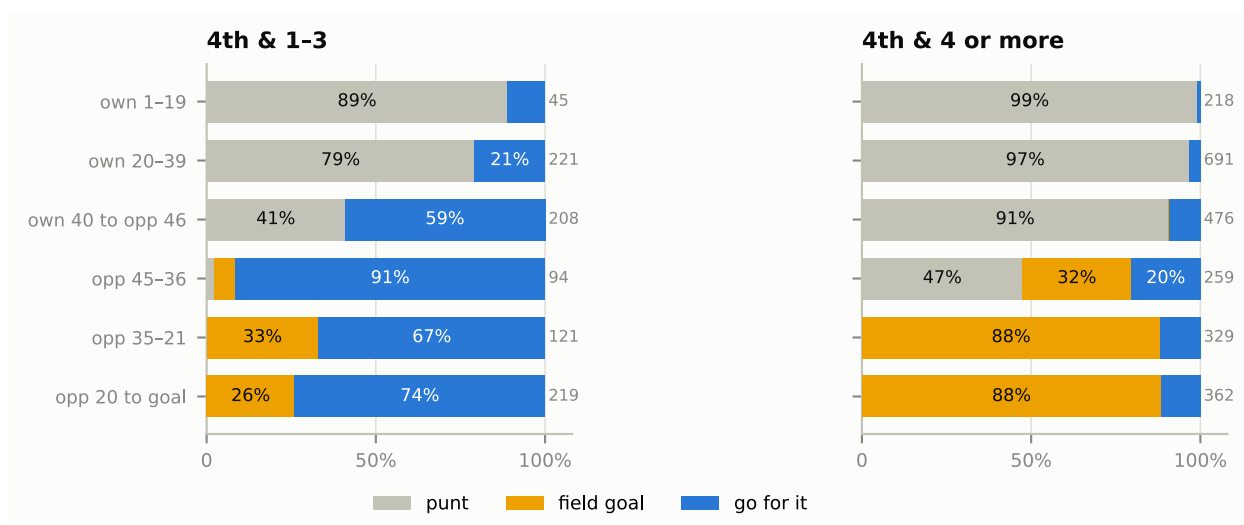


Figure 6: What NFL teams chose on fourth down, 2025 regular season, by where the ball was and how far they needed to go (excluding the last two minutes of each half). Deep in their own territory, teams punt; inside the opponent's 35, they kick field goals. Between them is a band near midfield where short fourth downs are now usually played out. The small grey numbers are the plays in each row. Source: nflverse play-by-play.

The left panel is the more interesting one. On fourth-and-short in your own end, teams still mostly punt: fail there and you've handed the opponent a short field. But from around your own 40 to the opponent's 36, a stretch where a punt gains little (the ball may bounce into the end zone) and a field goal is too long, teams now go for it on most short fourth downs. They didn't always. On fourth-and-3-or-less in close games, NFL teams went for it 25% of the time in 2016 and 53% in 2025.⁷ The change came from the analytics staffs you met in *What a Play Really Is*, armed with exactly the kind of numbers this chapter builds toward: a way

⁶NPR, "Say goodbye to chain crews: The NFL will use camera technology to measure 1st downs" (Apr. 1, 2025): six Hawk-Eye cameras per stadium, officials still spot the ball, the chain crew kept as a backup, measurements in about 30 seconds: <https://www.npr.org/2025/04/01/nx-s1-5347990/nfl-first-downs-measurements-hawk-eye> ; CBS Sports on the 2025 Hall of Fame Game debut (six 8K cameras, about 30 seconds per measurement): <https://www.cbssports.com/nfl/news/no-chain-no-problem-nfl-debuts-new-virtual-measurement-system-during-hall-of-fame-game> ; Sony/Hawk-Eye announcement (a virtual recreation shown on TV broadcasts and stadium video boards): <https://sony.mediaroom.com/2025-04-01-Sonys-Hawk-Eye-Innovations-Selected-by-the-NFL-to-Revolutionize-Line-to-Gain-Measurements> ; ESPN, 2025 rules round-up: https://www.espn.com/nfl/story/_/id/46088927/nfl-rules-changes-tush-push-chain-gang-kickoffs-celebrations

⁷Course calculation from nflverse play-by-play, regular seasons: fourth downs with 3 or fewer yards to go and the offense's win probability between 10% and 90%, counting runs and passes as "go for it" against punts and field-goal attempts.

to compare “a few yards of field” with “a 58% chance to keep the ball” (the fourth-and-2 conversion rate since 2016). The fourth-down decision gets its full treatment in Win Probability and Decision Analysis.

! Why it exists: the punt

The punt exists because the down system makes losing the ball inevitable for most drives, and because not all lost balls are equal. Giving the ball up at the opponent’s 30 instead of your own 30 is worth several points, as you’ll see when we price field position below. What it gives up is the chance, better than one in two on fourth-and-short, of keeping the drive alive. For most of NFL history coaches priced that chance too low and punted too often; the steady climb in go-for-it rates is the league correcting that.

Losing the ball: turnovers and the sack

A drive can also end without a kick. A **turnover** is any way of losing the ball to the other team without kicking it. There are three:

- An **interception**: a forward pass caught by a defender. The defense becomes the offense on the spot and can run the ball back, sometimes all the way for a touchdown (a “pick-six”).
- A **fumble**: the ball carrier loses control of the ball before he is down. It’s a live ball, and whoever recovers it has it, so a fumble is only a turnover if the defense comes up with it. (Who ends up on a loose ball is close to random, which matters for how much to trust a team’s turnover record; see Expected Points, EPA, and Success Rate.)
- A **turnover on downs**, from the last section: failing on fourth down.

The 2025 regular season had 380 interceptions and 248 lost fumbles, about 2.3 per game between the two teams. They are rare, and they are enormous. In the 204 games in which one team gave the ball away more often than the other, the team with fewer giveaways won 79% of the time.⁸

Why does a turnover hurt so much more than a bad play? Because it costs twice. You lose whatever your own possession was worth, and you hand the opponent a possession that starts wherever the turnover happened, often in good field position because your offense had been moving toward their end. A turnover doesn’t just stop your drive; it starts theirs. We’ll put a number on that below.

The sack

A **sack** is the quarterback being tackled behind the line of scrimmage before he can throw. It isn’t a turnover (the offense keeps the ball), but it’s the next worst thing on a passing play. It costs yards, about six or seven on average, and it costs a down, and those combine: second-and-10 becomes third-and-17, and the chart of third-down conversion rates shows what that does to the drive’s chances. A sack also carries a fumble risk, because the quarterback is often hit from behind while holding the ball loosely. There were 1,287 sacks in the 2025 regular season, on about 6.5% of dropbacks, and the Browns’ Myles Garrett set the single-season record with 23.⁹

In this book’s data a sack counts as a **pass play**, because the play called was a pass; the yards lost come off the passing game, not the running game. Who is to blame for a sack (the blockers, the quarterback holding the ball, receivers who couldn’t get open) is a long conversation that starts in Pass Protection.

Field position: the currency of the game

Put the pieces together. Every drive starts somewhere, and where it starts changes what it’s likely to produce. That’s **field position**: where the ball is, treated as an asset. A drive that starts at the opponent’s 40 is worth far more than one that starts at your own 10, before anyone has run a play.

The broadcast divides the field in half at the 50, so “own 30” means 30 yards from your own goal line and “opp 30” (or “the Chiefs’ 30”) means 30 yards from a touchdown. Coaches carve it up more finely:

⁸Course calculation from nflverse play-by-play, 2025 regular season: interceptions plus lost fumbles per team per game; games with an equal number of giveaways, and ties, are left out.

⁹NFL.com, “Browns DE Myles Garrett reaches 23 sacks, sets new NFL single-season record” (Jan. 2026): <https://www.nfl.com/news/browns-de-myles-garrett-reaches-23-sacks-sets-new-nfl-single-season-record>. Sack and dropback counts are a course calculation from nflverse play-by-play.

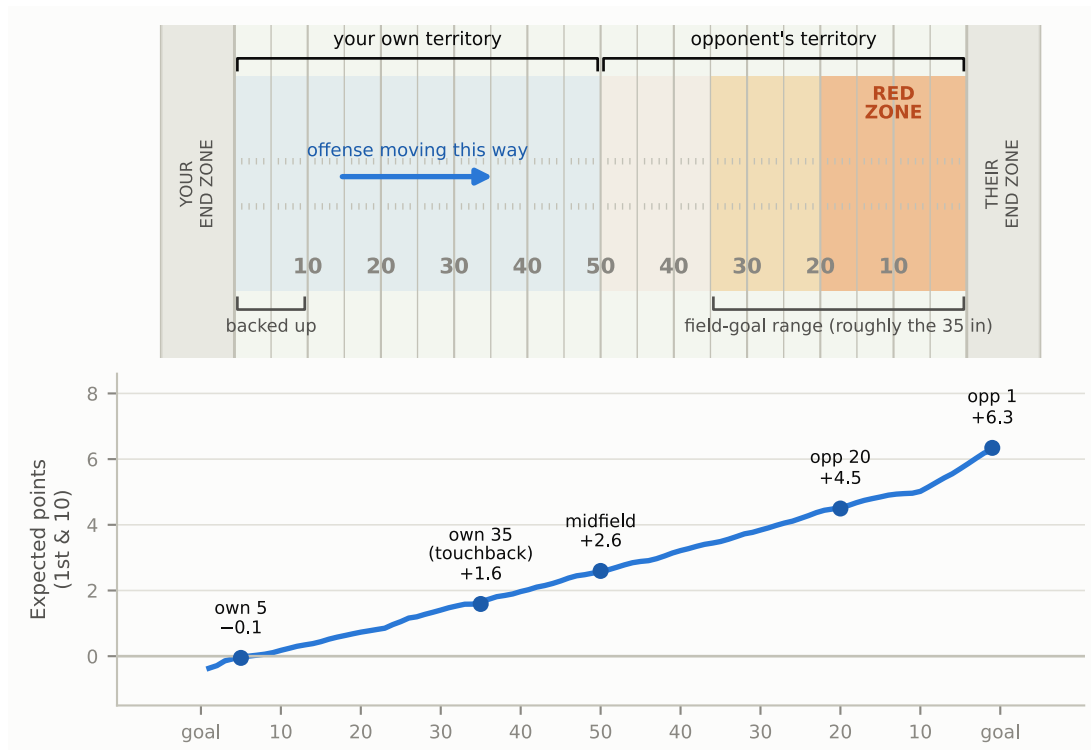


Figure 7: The field as a map of value, offense moving left to right. Top: the zones coaches talk about. ‘Backed up’ is inside your own 10; field-goal range starts somewhere around the opponent’s 35 (it depends on the kicker and the weather); the red zone is inside the opponent’s 20. Bottom: what a first-and-10 at each yard line is worth, in expected points (nflverse model, averaged over 2021-2025 snaps). Every yard is worth something, and the ladder has no flat steps.

The top half is the vocabulary. **Own territory** and **opponent’s territory** split at the 50. Inside your own 10 you are backed up: a sack or a holding penalty can mean a safety, and even a good drive has to travel 90-plus yards. Somewhere around the opponent’s 35, a drive enters field-goal range: from there a kick is roughly 52 yards or shorter, because the kicker stands about seven yards behind the line and the goalposts sit at the back of the end zone, ten more yards on. (Add 17 to the yard line to get a field goal’s length.) Inside the opponent’s 20 is the **red zone**, the part of the field where drives are expected to produce points and the only question is how many: seven or three, the central tension from the last chapter. The red zone plays differently from the rest of the field because it’s short: with the end zone only 20 yards away, the defense has less ground to protect and can crowd the receivers, and long passes over the top are no longer possible. The Red Zone, Goal Line, and Short Yardage shows how both sides adjust.

The bottom half is the new idea, and the rest of the chapter builds on it.

Expected points: what a situation is worth

If field position is a currency, we need prices. Here is the idea that supplies them.

Take every first-and-10 from your own 25 in recent NFL seasons, thousands of them. For each one, look ahead to the **next score** in the game: maybe this same drive ended in a touchdown (+7, counting the extra point), maybe it stalled and the offense kicked a field goal (+3), maybe it punted and the *other* team scored a field goal on its next drive (–3, because it’s the opponent’s points), maybe no one scored before halftime (0). Average all of those outcomes. The result is the **expected points (EP)** of first-and-10 at your own 25: what that situation is worth, on average, to the team with the ball.

Do the same for every combination of down, distance and yard line and you get a price list for the whole game. The bottom of the field map shows its most important row, first-and-10, and a few numbers are worth carrying in your head:

- First-and-10 at your **own 5** is worth about –0.1. Pinned that deep, the next score is about as likely to be the opponent’s as yours.
- At your **own 35**, where a kickoff that lands in the end zone has been spotted since 2025,¹⁰ it’s worth about +1.6.
- At **midfield**, about +2.6.
- At the **opponent’s 20**, the edge of the red zone, about +4.5.

¹⁰NFL.com, “NFL owners vote to make dynamic kickoff permanent, adjust ball spot on touchbacks to 35-yard line” (2025): <https://www.nfl.com/news/nfl-owners-vote-to-make-dynamic-kickoff-permanent-adjust-ball-spot-on-touchbacks-to-35-yard-line> . Kickoffs are covered in chapter 09-02.

- First-and-goal at the 1, about +6.3. Not quite 7, because even from the 1 a drive sometimes ends in a field goal or a turnover.

The down and distance move the price as much as the yard line does. At midfield with 10 to go:

Table 1: Expected points at midfield (the 48 to the 52) with 10 yards to go, by down: nflverse model values averaged over 2021-2025 snaps. The yard line is the same in every row; only the down changes.

Situation	Expected points
1st & 10 at midfield	+2.6
2nd & 10 at midfield	+2.0
3rd & 10 at midfield	+1.3
4th & 10 at midfield	−0.1

Each down you burn without gaining ground costs a chunk of value, and fourth-and-10 is worth about nothing: you’re probably punting, and the next score is as likely to be theirs as yours.

Two things about EP are worth saying plainly, because they’re easy to get wrong.

EP describes an average team. The model knows the down, distance and yard line (and, in its full form, a few more things such as the time left), but nothing about who is playing. It answers “what is this situation worth to a typical NFL offense against a typical defense?” That’s a feature, not a bug: it gives every play the same yardstick, so you can ask how much better than typical a team or a call was.

You don’t need to build it to use it. The values in this book come from the open-source nflverse model, fitted on decades of play-by-play. How such a model is built, where it came from (a Bengals quarterback named Virgil Carter published the first version in 1971¹¹), and how far to trust it are the subject of Expected Points, EPA, and Success Rate. For now, treat EP as the price tag on a situation.

! Why it exists: one currency for every play

Before expected points, football had no common unit. Is a 4-yard run on first down better or worse than a 4-yard run on third-and-3? Is a punt that pins the opponent at their 5 worth more than a long field goal attempt? Yards can’t answer, because a yard near your own goal line and a yard near the opponent’s aren’t worth the same, and a yard short of the line to gain isn’t worth the same as a yard past it. Expected points convert everything (yards, downs, field position, turnovers, kicks) into one unit, points, so every decision can be compared on one scale. What it gives up is detail: it is an average, so it can’t know that *your* quarterback is great or that it’s snowing.

EPA: a scorecard for every play

If every situation has a price, every play has a profit or loss: the price after the play minus the price before it. That’s **EPA, expected points added**:

$$\text{EPA} = \text{EP after the play} - \text{EP before the play}$$

Two rules cover the special cases:

- **If the play scores**, “EP after” is the points scored (7 for a touchdown, since the extra point is nearly automatic; 3 for a field goal).
- **If the ball changes hands**, “EP after” is the *negative* of what the new offense’s situation is worth, because their gain is your loss.

And one definition built on top of it. A play with positive EPA left the offense better off than before the snap; call that a **success**. A team’s **success rate** is simply the share of its plays with positive EPA. EPA per play measures *how much* value an offense creates on average; success rate measures *how often* it moves forward. You’ll see both on almost every chart in this book.

That’s all the machinery. Let’s use it.

¹¹Virgil Carter and Robert E. Machol, “Technical Note: Operations Research on Football,” *Operations Research* 19, no. 2 (1971): 541–544, <https://doi.org/10.1287/opre.19.2.541>. The full story is in chapter 13-02.

One real drive, scored snap by snap

In the last chapter you located the Super Bowl LIX scoring plays on the field. Here is the first one, built up snap by snap. Philadelphia got the ball at its own 31 midway through the first quarter, with the game still scoreless, and scored a touchdown eight snaps later.¹²

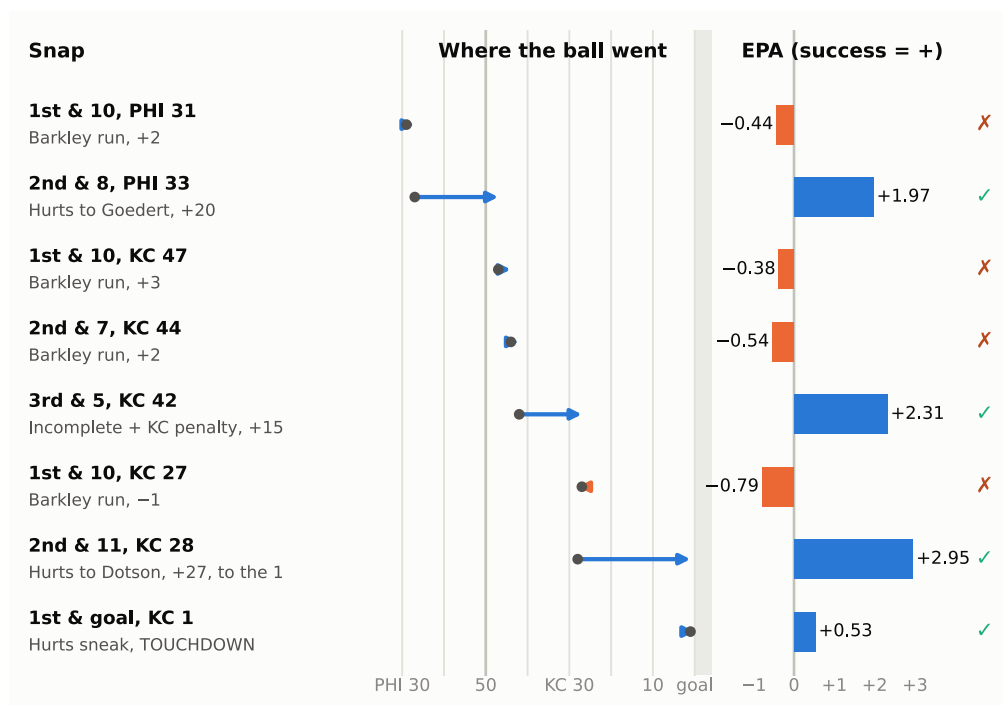


Figure 8: An EPA scorecard for the Eagles' first touchdown drive in Super Bowl LIX (February 9, 2025; Eagles 40, Chiefs 22). Each row is one snap: the situation before it, where the ball went (blue arrow = forward, orange = backward), and the play's EPA. Blue bars and ticks are successes (positive EPA). Four short runs all gained ground or nearly so and all scored negative; two completions and one defensive penalty carried the drive. Source: nflverse play-by-play (game 2024_22_KC_PHI).

Walk down the card.

Snap 1, Barkley run for 2. First-and-10 at the Eagles' 31 was worth +1.37 points. Two yards later it's second-and-8, and second-and-8 at the 33 is worth *less* than first-and-10 at the 31: the offense gained two yards but spent a down, and the down was worth more. EPA -0.44. This is the most important lesson on the card: **a gain can be a loss**. On first-and-10, a run needs roughly four or five yards just to break even.

Snap 2, Hurts to Goedert for 20. A first down *and* twenty yards of field. EPA +1.97, the drive's first big credit.

Snaps 3 and 4, two more short Barkley runs. Three yards, then two, leaving third-and-5. Each is a small debit.

Snap 5, incomplete, but a penalty. On third-and-5 the pass fell incomplete, which would have brought on the punt team. Instead Kansas City was flagged for unnecessary roughness: fifteen yards and an automatic first down. The play-by-play scores penalties like any other snap, by the situation before and after, and this one was worth +2.31, the difference between third-and-5 at the Kansas City 42 and first-and-10 at the 27.

Snaps 6 and 7. A one-yard loss (-0.79), then the play of the drive: a 27-yard completion to Jahan Dotson, first ruled a touchdown and changed on review to a catch at the 1. EPA +2.95: from second-and-11 to first-and-goal at the 1 is a huge jump in price.

Snap 8, the touchdown. Jalen Hurts pushed in from the 1 on the Eagles' famous quarterback sneak (the tush push). First-and-goal at the 1 was already worth +6.47; scoring the 7 added only the rest, +0.53. The hard work of this touchdown was done on the play before.

Now add up the column. The eight EPAs sum to +5.63, and the drive began with +1.37 of expected points in hand. Together they make 7.00: the seven points the drive actually scored. That is not a coincidence. EPA is bookkeeping: it divides a drive's points among the plays that produced them, and the total always matches what happened. Success rate on the drive: 4 of 8.

¹²Super Bowl LIX, February 9, 2025, Caesars Superdome, New Orleans: Eagles 40, Chiefs 22 (CBS News, <https://www.cbsnews.com/amp/news/nfl-mvp-super-bowl-jalen-hurts-eagles>). Play-by-play, EP and EPA from nflverse (game 2024_22_KC_PHI, the Eagles' drive starting at 9:40 of the first quarter, and the interception at 14:21 of the second).

Common misconception

“Every gain is a good play, and every loss is a bad one.” The scorecard says otherwise in both directions. Saquon Barkley’s three short runs all gained ground and all scored negative, because each used up a down without gaining enough to pay for it. (That says little about Barkley, who ran for over 2,000 yards that season; four plays are a sample of nothing. It says a lot about how EPA prices first-and-second-down runs.) Meanwhile an 8-yard catch on third-and-12 scores negative too, because it leaves fourth-and-4 and a punt. Yardage tells you what happened; EPA tells you whether it helped.

Turnovers on the scorecard

Run the same arithmetic on a turnover and you see why they decide games. Early in the second quarter the Eagles faced third-and-10 at the Kansas City 30, a situation worth about +2.83 points to them. Jalen Hurts threw deep, and Kansas City’s Bryan Cook intercepted at his own 2. Now the “after” situation is the Chiefs’ first-and-10 at their 2, worth slightly *less than zero* to the Chiefs (pinned deep, they’re as likely to give up the next score as make it), so slightly more than zero to the Eagles. EPA for the throw: -2.51. The Eagles lost a near-certain field goal attempt in one snap. Had Cook returned it to midfield, the cost would have been bigger still, because the “after” would have been a Kansas City first down in good field position. That’s the “turnovers cost twice” idea from earlier, as one number.

Drives and the red zone

Step back from single plays to the unit coaches plan around. A **drive** is one team’s unbroken run of plays with the ball, from the moment it gets it (a kickoff, a punt, a turnover) until it scores, kicks it away, turns it over, or the half ends. It’s the same thing as the possession from the last chapter, seen from the inside.

In 2025 each team had about 10.1 drives per game, and the typical drive started near its own 29-yard line. They ended like this: a touchdown on 23%, a field goal on 16%, a punt on 33%, a turnover of some kind (including on downs) on 16%, and the rest at the end of a half or with a missed kick.¹³ A third of all drives end in a punt; roughly four in ten end in points.

Those numbers explain why the expected-points ladder reads only about +1.4 at a typical drive start, rather than near zero or near 7. A typical drive is a roughly 40% shot at points, more often seven than three, with a real chance of handing the other team good field position along the way.

The red zone is where drives cash in. Drives that reach it end in points most of the time, and the fight is over which kind: the last chapter showed how often it’s seven and how often three. Red-zone efficiency, the share of red-zone trips that end in touchdowns, is one of the numbers broadcasters quote most, and you can now see why it matters: on the field map, the gap between a first-and-10 at the opponent’s 20 and a touchdown is about 2.5 points, and a field goal there captures only some of it.

¹³Course calculation from nflverse play-by-play (`fixed_drive_result`), 2025 regular season, all drives; the typical start is the median first snap of a drive.

Two of the most famous stops in Super Bowl history look alike: an offense a few yards from a tying or winning touchdown, a defense that holds. The down-and-distance graphic says they were completely different situations.

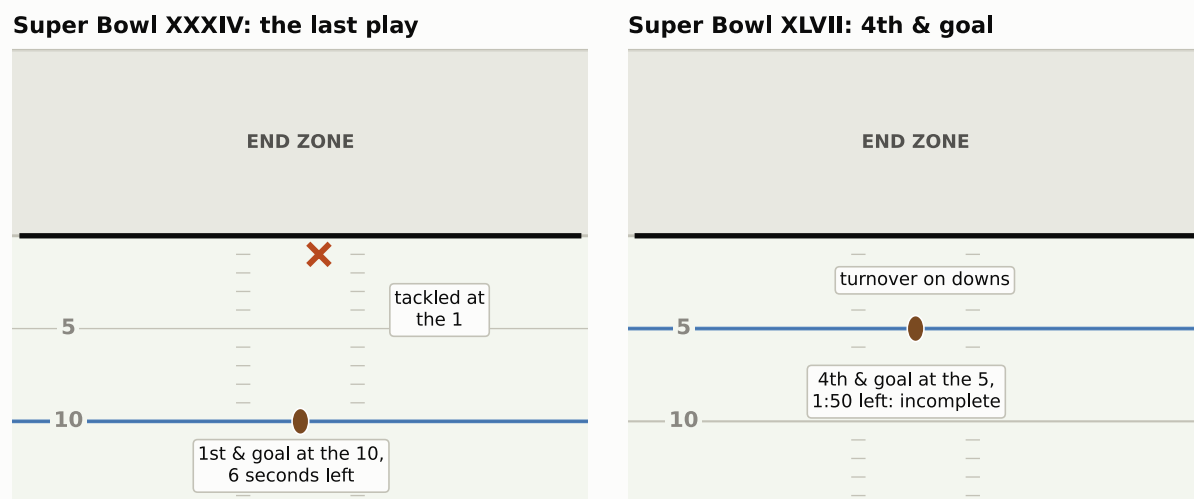


Figure 9: Two goal-line stands, located on the field (spots only; the plays themselves are not reconstructed). Left: Super Bowl XXXIV, the Titans' last snap from the Rams' 10 with 6 seconds left; Kevin Dyson was tackled at the 1 as time expired. Right: Super Bowl XLVII, the 49ers' fourth-and-goal from the Ravens' 5 with 1:50 left; the pass fell incomplete and Baltimore took over on downs.

Super Bowl XXXIV, January 30, 2000. The Rams led the Titans 23–16. Tennessee drove from its own 12 and, after a completion and a timeout, had first-and-goal at the Rams' 10 with six seconds left: time for one play. Steve McNair hit Kevin Dyson on a slant inside the 5, and Rams linebacker Mike Jones wrapped him up; Dyson stretched the ball toward the goal line and was down at the 1 as time expired.¹⁴ “One Yard Short.” Notice the down: *first* and goal. This was not a fourth-down stop. The Titans had downs to spare and no time to use them. When the clock runs out, the down and distance stop mattering (how the clock interacts with the downs is in Clock and Game Management).

Super Bowl XLVII, February 3, 2013. The Ravens led the 49ers 34–29. San Francisco reached first-and-goal at the Baltimore 7 with time for a full series. A run gained two, and three straight passes to Michael Crabtree fell incomplete, the last on fourth down with 1:50 left, under pressure and with contact between Crabtree and Ravens cornerback Jimmy Smith that drew no flag. Turnover on downs: Baltimore's ball at its own 5 with 1:46 to play.¹⁵

Then watch what Baltimore did with the worst field position in football. It ran three plays to burn the clock, and with 12 seconds left punter Sam Koch, rather than punting from his own end zone, ran around in it for eight seconds before stepping out: an intentional safety. Two points to the 49ers (34–31), but the clock was nearly gone, and the free kick that follows a safety let Baltimore kick from its 20 instead of punting from inside its own end zone. The 49ers' return came up short, and the Ravens won.

What to notice: both defenses won, but on different levers. In XXXIV the defense beat the *clock*, with downs left over. In XLVII it beat the *downs*: four tries from the 7 and no touchdown, which is the cleanest definition of a goal-line stand. And Baltimore's safety is field position as currency in its purest form: the Ravens *paid* two points to buy distance and time.

¹⁴“Super Bowl XXXIV,” Wikipedia (drive from the Titans' 12; first-and-goal at the 10 with six seconds left; Dyson tackled by Mike Jones short of the goal line): https://en.wikipedia.org/wiki/Super_Bowl_XXXIV ; box score at Pro Football Reference: <https://www.pro-football-reference.com/boxscores/200001300ti.htm> ; ESPN.com game story (Jan. 30, 2000; Titans' final timeout with six seconds left): <https://www.espn.com/nfl/playoffs99/sb/s/frozen/0130ratio.html> . The drive is also in nflverse play-by-play (game 1999_21_STL_TEN), whose clock log puts the last timeout at 0:05.

¹⁵“Super Bowl XLVII,” Wikipedia (first-and-goal at the 7; LaMichael James 2-yard run; three incompletions; turnover on downs with 1:46 left; Sam Koch's intentional safety with 12 seconds left): https://en.wikipedia.org/wiki/Super_Bowl_XLVII . Sequence and clock checked against nflverse play-by-play (game 2012_21_BAL_SF): fourth-down snap at 1:50, Baltimore ball at 1:46; Rice, Pierce and Leach runs; Koch out of the end zone at 0:04 after a 0:12 snap; Koch's free kick from the 20 returned 31 yards by Ted Ginn as time expired.

💡 Watch for it on Sunday: the down-and-distance forecaster

This is your baseline. For one half of one game, before every snap:

- **Read the graphic and say the bucket out loud:** “second and long,” “third and short.”
- **Predict run or pass from the down and distance alone.** Ignore the formation for now (you don’t have the tools yet). Mark your guess and the result on a scrap of paper: two columns, R or P.
- **Note anything that broke the rule.** A pass on third-and-1, a run on third-and-8: what was the score and the clock? Usually you’ll find a reason.
- **At halftime, compute your hit rate.**

What to expect: if you had simply guessed “pass” on every neutral-situation snap in 2025, you’d have been right 58% of the time. The simplest sensible rule (“pass on first-and-more-than-10, second-and-7-plus, and third-or-fourth-and-4-plus; otherwise run”) was right 65% of the time. If you’re near that, you’re reading the graphic like the league average. Keep the sheet: How to Watch a Broadcast Like You Mean It turns it into a formal prediction log, and every later chapter adds a clue that should push your number up.

Predict the play

💡 Predict the play 1: third-and-1 near midfield

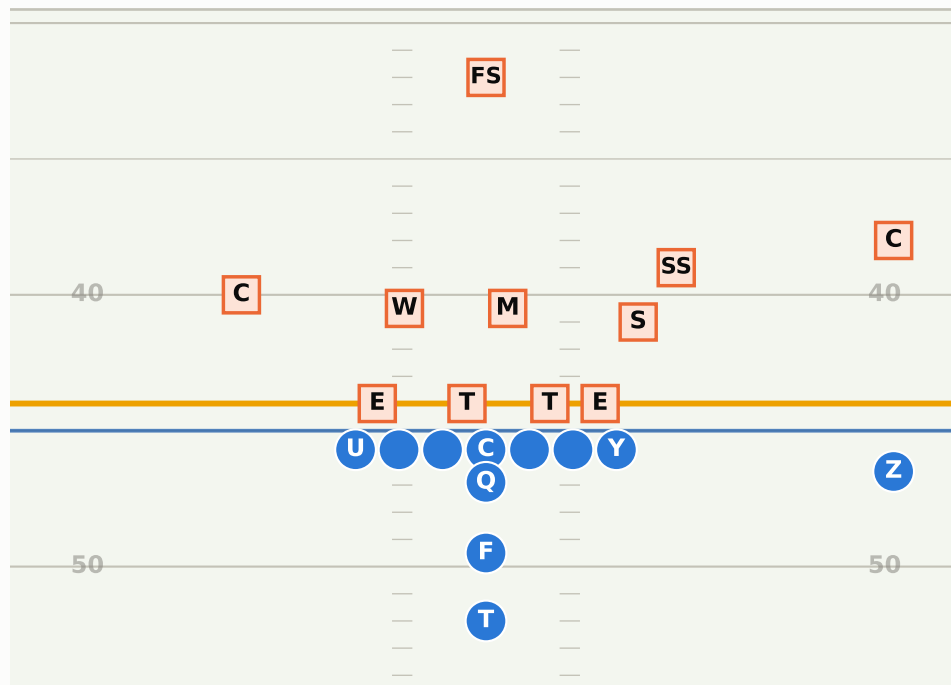


Figure 10: Third-and-1 at the opponent’s 45, second quarter, tied. The offense brings in two tight ends (Y and U) and a fullback (F) and lines up with the quarterback under center. The defense answers with its bigger players: four linemen, three linebackers, and a safety (SS) walked down near the line, with one safety deep.

Run or pass? And if they fail, what happens on fourth down?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: second-and-15 after a sack

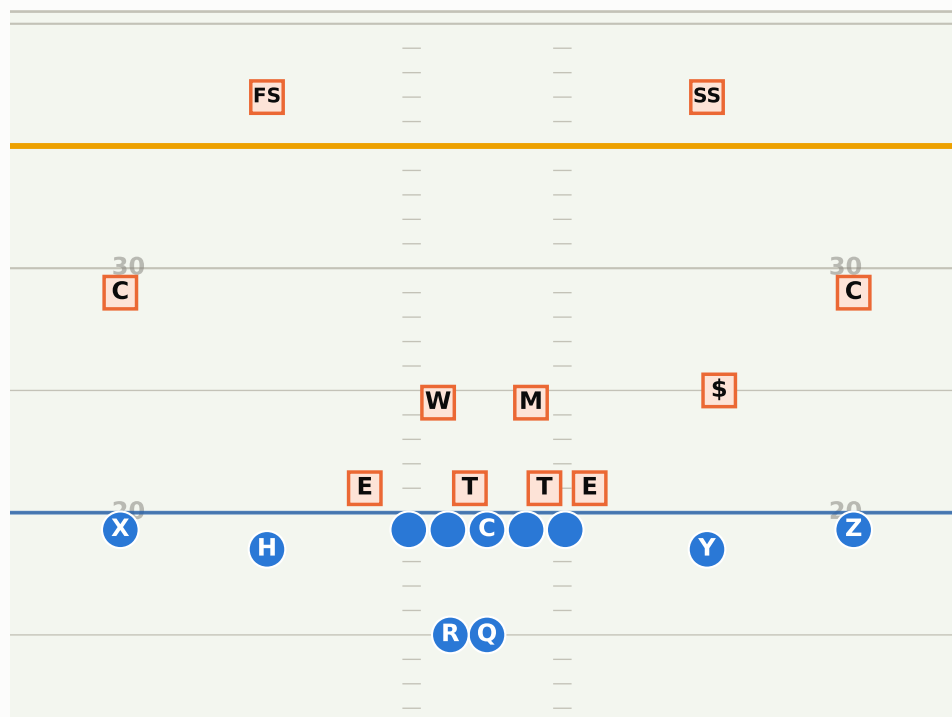


Figure 11: Second-and-15 at the offense's own 20, first quarter, after a 5-yard sack on first down. The offense is in the shotgun with four receivers spread out; the defense has five defensive backs, with both safeties (FS, SS) 17 yards deep and the cornerbacks 9 yards off their receivers. The line to gain (yellow) is the 35.

What is the offense likely to call, and what is the defense happy to give up?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: fourth-and-2 at the opponent's 38

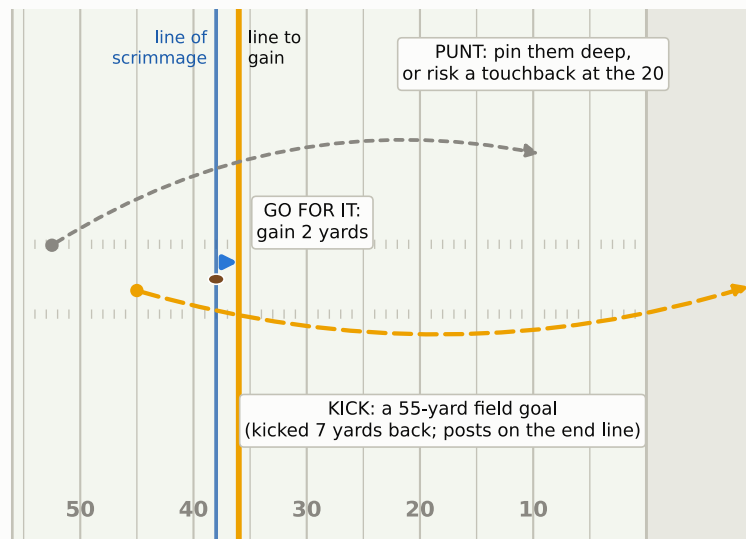


Figure 12: Fourth-and-2 at the opponent's 38, third quarter, down by 3. The offense's three options, drawn on the field: go for the 2 yards; kick a 55-yard field goal (snapped from 7 yards behind the line, with the goalposts at the back of the end zone); or punt, trying to pin the opponent deep without the ball bouncing into the end zone for a touchback.

Go, kick or punt? What did NFL teams usually do here in 2025, and why?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Four downs to go ten yards** is a bargain: a guarantee of four tries in exchange for an obligation to make progress. The down count and the line to gain reset only when the offense earns a first down; everything else on the field keeps moving.
- **The graphic predicts the play.** Short, medium and long change what the offense can do, so they change what it calls: first-and-10 is a toss-up, third-and-4-plus is almost always a pass, and every extra yard on third down costs a few percentage points of conversion.
- **The spot is a human judgment; the measurement is now a camera.** Officials spot the ball at its forward progress, the sticks show the series on the sideline, and since 2025 Hawk-Eye measures close calls, with the chains as backup.
- **Losing the ball comes in four kinds**, three of them turnovers (interception, lost fumble, turnover on downs) and one chosen (the punt, which trades a possession you'd probably lose for field). A turnover costs twice; a sack costs yards *and* a down.
- **Field position is currency, and expected points are its prices.** First-and-10 is worth about 0 at your own 5, 1.6 at your own 35, 2.6 at midfield, 4.5 at the opponent's 20 and 6.3 at the 1.
- **EPA is the change in price on one play, and success rate is the share of plays that raised it.** Gains can score as losses, and a drive's EPA always adds up to the points it actually produced.

Connections

Builds on: The Game in Fifteen Minutes: the field's markings, the five ways to score, possessions and the clock.

Sets up: The Twenty-Two names the players who fight over every one of these yards. What a Play Really Is shows how the down and distance organizes the coach's call sheet. How to Watch a Broadcast Like You Mean It turns this chapter's forecaster into a prediction log. Situational football in Part 10 is this chapter grown up: Early Downs and Field Zones, Third Down, The Red Zone, Goal Line, and Short Yardage and Clock and Game Management. The kicking game is in Field Goals, Extra Points, and the Punting Game, the measurement rules in Officiating, Replay, and the Rules of the Ball. Every EPA chart in the book (zone runs, pass concepts, blitzes) leans on the primer here; the model itself is opened up in Expected Points, EPA, and Success Rate, and fourth-down decisions in Win Probability and Decision Analysis.

Case studies: the tush push that finished our scorecard drive is the centerpiece of the Eagles' trenches and the tush push; the fourth-down aggression of the 2020s Lions is in the Lions under Ben Johnson and Dan Campbell.

i Go deeper

- Bob Carroll, Pete Palmer and John Thorn, *The Hidden Game of Football* (1988): the book that taught a generation of fans to think of field position in points.
- The nflfastR documentation (<https://www.nflfastR.com/>) and the Open Source Football site (<https://www.opensourcefootball.com/>): the expected-points model behind every number in this chapter, explained by the people who maintain it.
- Pat Kirwan, *Take Your Eye Off the Ball* (2nd ed., 2015): a former NFL coach on how down and distance shape what both sides do.
- Brian Burke's essays at ESPN and his old Advanced NFL Stats archive: the clearest popular writing on fourth-down decisions.

Answers to Predict the play

Answer 1.

Probably a run, but it's closer than it looks, and fourth down is part of the plan.

Short yardage is the one place on the grid where a run is the normal call: on third-and-1 to 3 in 2025, offenses ran 54% of the time, and heavier lineups like this one (extra tight ends and a fullback, the subject of Personnel Groupings) lean further toward the run. The defense has answered with size and an extra man near the line. Third-and-1 is converted about 69% of the time across the last decade, the best odds on the third-down chart, and quarterback sneaks convert more often still (The Red Zone, Goal Line, and Short Yardage).

But a defense that loads up like this is betting on the run, and that's exactly when a play-action pass (fake the run, throw behind the crowd) is most dangerous. That's why the pass share on third-and-short is still near half.

The bigger idea: at the opponent's 45, a miss on third down is unlikely to bring on the punt team. On fourth-and-1-to-3 between the opponent's 45 and 36 in 2025, teams went for it 91% of the time. Many offenses treat this third down as the first of **two** tries, which is why they can afford a run that might gain only a yard. Coaches call this four-down territory.

Answer 2.

A pass, most likely a short or medium one, and the defense is content to let it be caught in front of it.

Second-and-very-long is pass territory: offenses threw 82% of the time on second-and-11-plus in neutral situations in 2025, and 85% of the time on second-and-13-to-17 deep in their own end (2021–2025). The sack did the damage: it cost five yards and a down, and a third-and-long is now likely.

Look at the defense. Two safeties 17 yards deep and cornerbacks nine yards off are a defense saying “we won't let you get it all back at once.” It's willing to give up a 7-yard completion, because 7 yards still leaves third-and-8, converted only about a third of the time. What it can't afford is the 20-yard play that erases the sack in one snap.

EPA adds a twist. The price before this snap is already low: the sack spent it. So the bar for a “success” is lower than it was on first down. On second-and-13-to-17 near their own 20 (2021–2025), gains of about 6 yards or more were positive on average, because they set up a third down the offense can realistically convert. EPA always grades a play against the situation it started from, not against the original first-and-10.

A good counter you'll sometimes see: a **draw**, a run disguised as a pass, against a defense that has spread out to defend the pass. It rarely converts, but it can turn third-and-15 into a manageable third-and-8.

Answer 3.

Most teams now go for it. On fourth-and-1-to-3 between the opponent's 45 and 36 in 2025, offenses went for it 91% of the time. Price the three options with this chapter's tools:

- **Go for it.** Fourth-and-2 has been converted about 58% of the time (2016–2025). Make it and you have a first down around the opponent's 35, worth about +3.5 expected points. Miss it and the opponent takes over at its own 38, a decent but not disastrous start for them.
- **Kick.** A 55-yarder is no longer a prayer: NFL kickers made 70% of attempts from 53 to 57 yards in 2023–2025 (371 kicks). But a miss hands the opponent the ball at the spot of the kick, around its own 45, and a make is “only” 3.
- **Punt.** From the 38 there are only 38 yards to work with. A good punt pins the opponent inside its 10; a slightly long one bounces into the end zone for a touchback at the 20, which means the punt bought you 18 yards for your whole possession.

Field position is the reason the punt loses here: there isn't enough field left to buy. That's the band in the middle of the fourth-down chart where the bars turned blue. Being down by 3 nudges toward the field goal, which would tie the game; in Win Probability and Decision Analysis you'll see how the score and clock change the answer, and why most models still say go.